



MOTOROLA
Semiconductors

BOX 20912, PHOENIX, ARIZONA 85036

MC6820

Advance Information

PERIPHERAL INTERFACE ADAPTER (PIA)

The MC6820 Peripheral Interface Adapter provides the universal means of interfacing peripheral equipment to the MC6800 Microprocessing Unit (MPU). This device is capable of interfacing the MPU to peripherals through two 8-bit bidirectional peripheral data buses and four control lines. No external logic is required for interfacing to most peripheral devices.

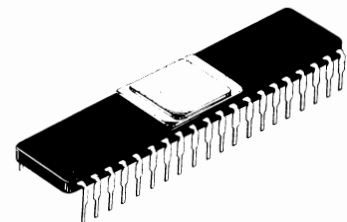
The functional configuration of the PIA is programmed by the MPU during system initialization. Each of the peripheral data lines can be programmed to act as an input or output, and each of the four control/interrupt lines may be programmed for one of several control modes. This allows a high degree of flexibility in the over-all operation of the interface.

- 8-Bit Bidirectional Data Bus for Communication with the MPU
- Two Bidirectional 8-Bit Buses for Interface to Peripherals
- Two Programmable Control Registers
- Two Programmable Data Direction Registers
- Four Individually-Controlled Interrupt Input Lines; Two Usable as Peripheral Control Outputs
- Handshake Control Logic for Input and Output Peripheral Operation
- High-Impedance 3-State and Direct Transistor Drive Peripheral Lines
- Program Controlled Interrupt and Interrupt Disable Capability
- CMOS Compatible Peripheral Lines

MOS

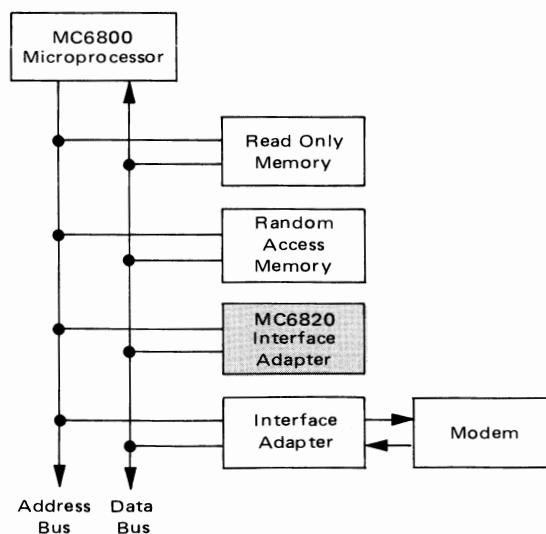
(N-CHANNEL, SILICON-GATE)

**PERIPHERAL INTERFACE
ADAPTER**

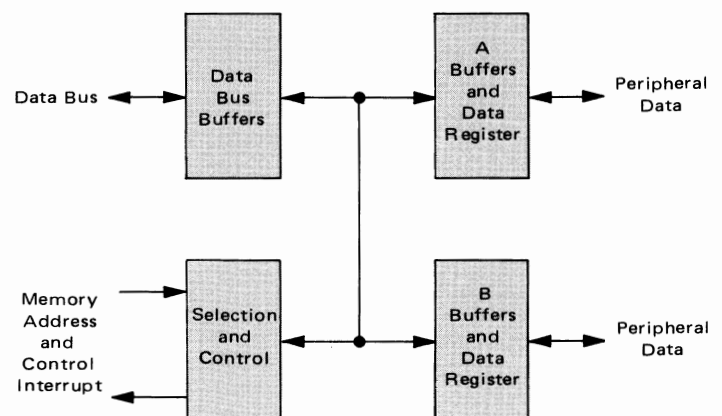


CERAMIC PACKAGE
CASE 699

**M6800 MICROCOMPUTER FAMILY
BLOCK DIAGRAM**



**MC6820 PERIPHERAL INTERFACE ADAPTER
BLOCK DIAGRAM**



MAXIMUM RATINGS

Rating	Symbol	Value	Unit
Supply Voltage	V_{CC}	-0.3 to +7.0	Vdc
Input Voltage	V_{in}	-0.3 to +7.0	Vdc
Operating Temperature Range	T_A	0 to +70	°C
Storage Temperature Range	T_{stg}	-55 to +150	°C

This device contains circuitry to protect the inputs against damage due to high static voltages or electric fields; however, it is advised that normal precautions be taken to avoid application of any voltage higher than maximum rated voltages to this high-impedance circuit.

ELECTRICAL CHARACTERISTICS ($V_{CC} = 5.0 \text{ V} \pm 5\%$, $V_{SS} = 0$, $T_A = 25^\circ\text{C}$ unless otherwise noted.)

Characteristic	Symbol	Min	Typ	Max	Unit
Input High Voltage (Normal Operating Levels)	V_{IH}	$V_{SS} + 2.4$	—	V_{CC}	Vdc
Input Low Voltage (Normal Operating Levels)	V_{IL}	$V_{SS} - 0.3$	—	$V_{SS} + 0.4$	Vdc
Input High Threshold Voltage	V_{IHT}	$V_{SS} + 2.0$	—	—	Vdc
Input Low Threshold Voltage	V_{ILT}	—	—	$V_{SS} + 0.8$	Vdc
Input Leakage Current ($V_{in} = 0$ to 5.0 Vdc) R/W, Reset, RS0, RS1, CS0, CS1, CS2, CA1, CB1, Enable	I_{in}	—	1.0	2.5	μAdc
Three-State (Off State) Input Current ($V_{in} = 0.4$ to 2.4 Vdc, $V_{CC} = \text{max}$)	I_{TSI}	—	2.0	10	μAdc
Input High Current ($V_{IH} = 2.4 \text{ Vdc}$)	I_{IH}	-100	-250	—	μAdc
Input Low Current ($V_{IL} = 0.4 \text{ Vdc}$)	I_{IL}	—	-1.0	-1.6	mA_{dc}
Output High Voltage ($V_{CC} = \text{min}$, $I_{Load} = -100 \mu\text{Adc}$, Enable Pulse Width < 25 μs)	V_{OH}	$V_{SS} + 2.4$	—	—	Vdc
Output Low Voltage ($V_{CC} = \text{min}$, $I_{Load} = 1.6 \text{ mA}_{dc}$)	V_{OL}	—	—	$V_{SS} + 0.4$	Vdc
Output High Current (Sourcing) ($V_{OH} = 2.4 \text{ Vdc}$) ($V_O = 1.5 \text{ Vdc}$, the current for driving other than TTL, e.g., Darlington Base)	I_{OH}	-100 -1.0	-1000 -2.5	—	μAdc mA_{dc}
Output Low Current (Sinking) ($V_{OL} = 0.4 \text{ Vdc}$)	I_{OL}	1.6	—	—	mA_{dc}
Output Leakage Current (Off State)	I_{RQA}, I_{RQB}	—	1.0	10	μAdc
Power Dissipation	P_D	—	300	600	mW
Input Capacitance ($V_{in} = 0$, $T_A = 25^\circ\text{C}$, $f = 1.0 \text{ MHz}$)	C_{in}	—	—	10 7.0 20	pF
Output Capacitance ($V_{in} = 0$, $T_A = 25^\circ\text{C}$, $f = 1.0 \text{ MHz}$)	C_{out}	—	—	10	pF

READ TIMING CHARACTERISTICS (Figure 1, Loading = 30 pF and one TTL load for PA0-PA7, PB0-PB7, CA2, CB2 = 130 pF and one TTL load for D0-D7, IRQA, IRQB)

Characteristic	Symbol	Min	Typ	Max	Unit
Delay Time, Address valid to Enable positive transition	T_{AEW}	180	—	—	ns
Delay Time, Enable positive transition to Data valid on bus	T_{EDR}	—	—	395	ns
Peripheral Data Setup Time	T_{PDSU}	300	—	—	ns
Data Bus Hold Time	T_{HR}	10	—	—	ns
Delay Time, Enable negative transition to CA2 negative transition	T_{CA2}	—	—	1.0	μs
Delay Time, Enable negative transition to CA2 positive transition	T_{RS1}	—	—	1.0	μs
Rise and Fall Time for CA1 and CA2 input signals	t_r, t_f	—	—	1.0	μs
Delay Time from CA1 active transition to CA2 positive transition	T_{RS2}	—	—	2.0	μs
Rise and Fall Time for Enable input	t_{rE}, t_{fE}	—	—	25	ns

WRITE TIMING CHARACTERISTICS (Figure 2)

Characteristic	Symbol	Min	Typ	Max	Unit
Enable Pulse Width	T_E	0.470	—	25	μs
Delay Time, Address valid to Enable positive transition	T_{AEW}	180	—	—	ns
Delay Time, Data valid to Enable negative transition	T_{DSU}	300	—	—	ns
Delay Time, Read/Write negative transition to Enable positive transition	T_{WE}	130	—	—	ns
Data Bus Hold Time	T_{HW}	10	—	—	ns
Delay Time, Enable negative transition to Peripheral Data valid	T_{PDW}	—	—	1.0	μs
Delay Time, Enable negative transition to Peripheral Data Valid, CMOS ($V_{CC} = 30\%$)	T_{CMOS}	—	—	2.0	μs
Delay Time, Enable positive transition to CB2 negative transition	T_{CB2}	—	—	1.0	μs
Delay Time, Peripheral Data valid to CB2 negative transition	T_{DC}	0	—	1.5	μs
Delay Time, Enable positive transition to CB2 positive transition	T_{RS1}	—	—	1.0	μs
Rise and Fall Time for CB1 and CB2 input signals	t_r, t_f	—	—	1.0	μs
Delay Time, CB1 active transition to CB2 positive transition	T_{RS2}	—	—	2.0	μs

Note: Negative sign indicates outward current flow, positive sign indicates inward flow.



FIGURE 1 – READ TIMING CHARACTERISTICS

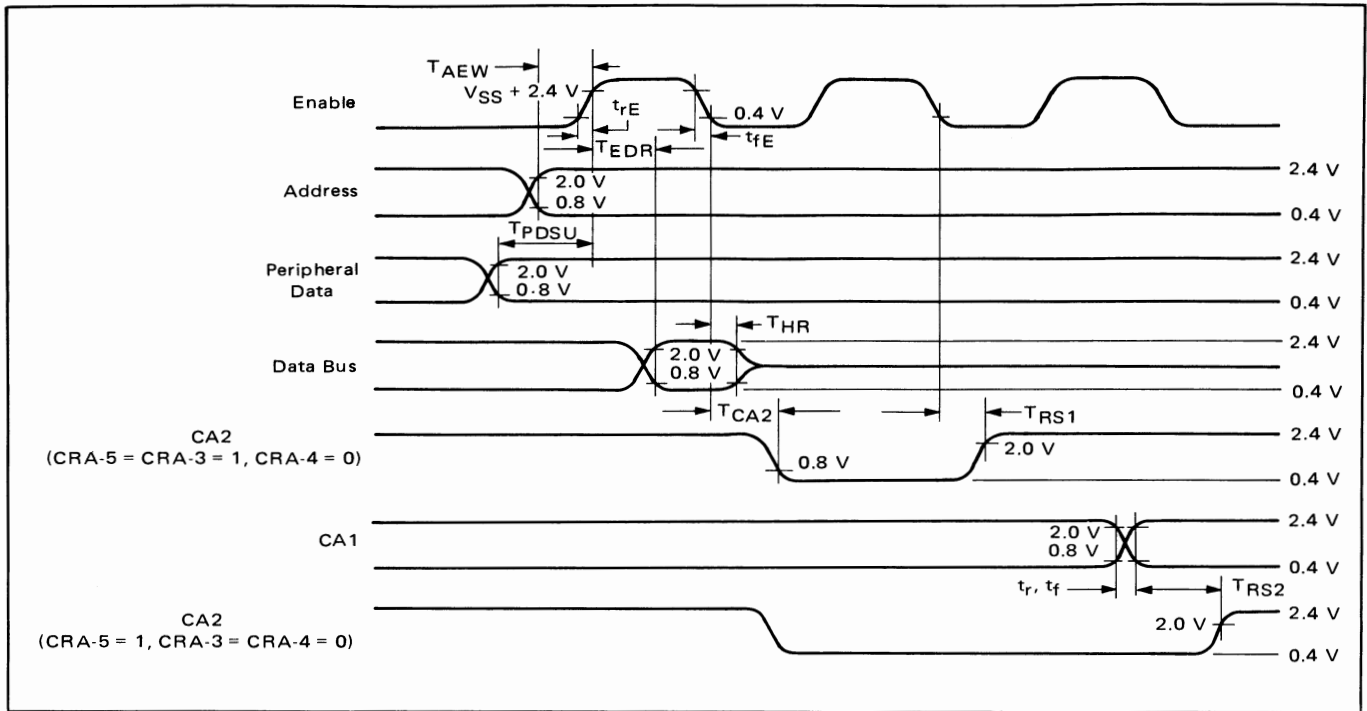
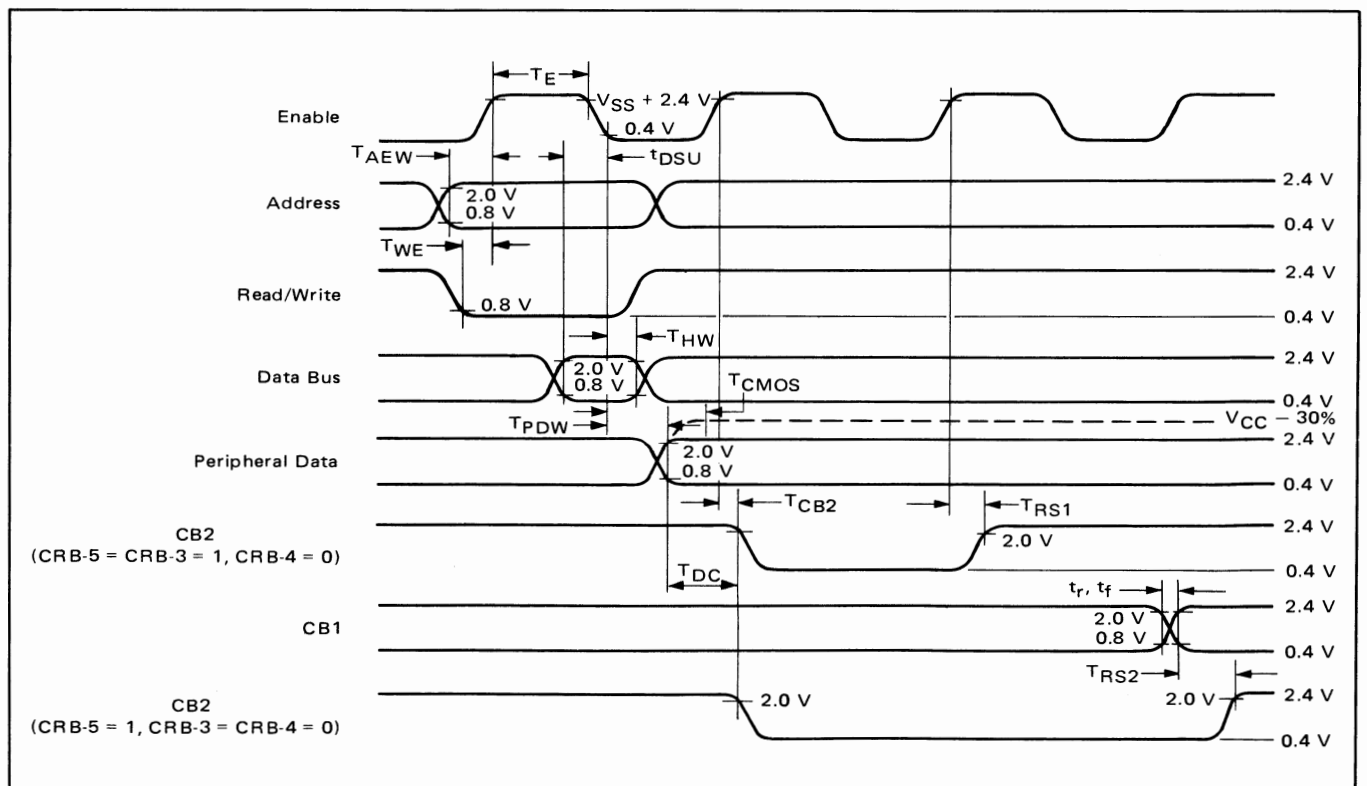
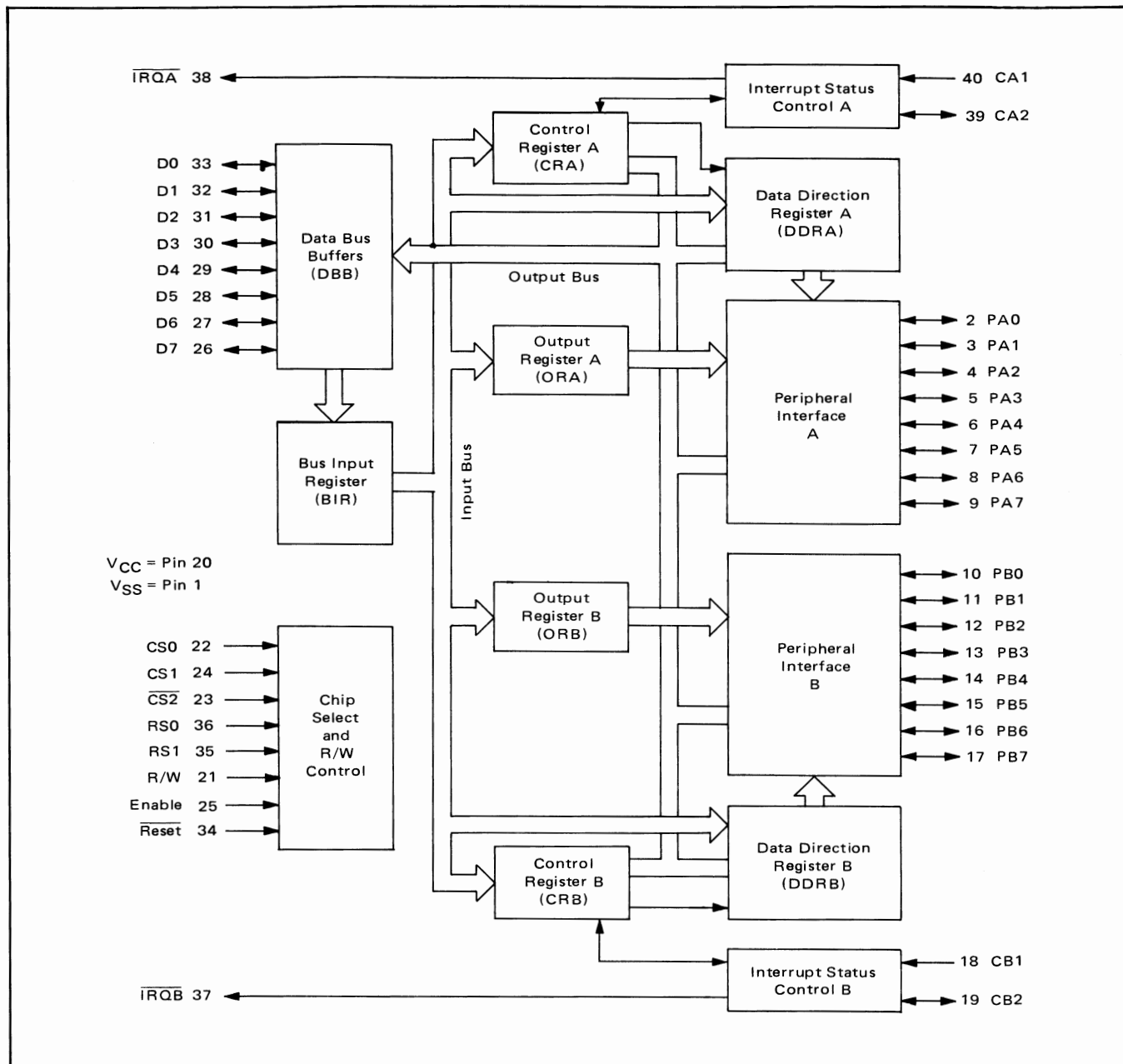


FIGURE 2 – WRITE TIMING CHARACTERISTICS



EXPANDED BLOCK DIAGRAM



PIA INTERFACE SIGNALS FOR MPU

The PIA interfaces to the MC6800 MPU with an eight-bit bi-directional data bus, three chip select lines, two register select lines, two interrupt request lines, read/write line, enable line and reset line. These signals, in conjunction with the MC6800 VMA output, permit the MPU to have complete control over the PIA. VMA may be utilized to gate the input signals to the PIA.

PIA Bi-Directional Data (D0-D7) — The bi-directional data lines (D0-D7) allow the transfer of data between the MPU and the PIA. The data bus output drivers are three-state devices that remain in the high-impedance (off) state except when the MPU performs a PIA read operation. The

Read/Write line is in the Read (high) state when the PIA is selected for a Read operation.

PIA Enable (E) — The enable pulse, E, is the only timing signal that is supplied to the PIA. Timing of all other signals is referenced to the leading and trailing edges of the E pulse. This signal will normally be a derivative of the MC6800 $\phi 2$ Clock.

The E pulse is used to condition the interrupt/control lines CA1, CA2, CB1, and CB2. At least one E pulse must occur from the inactive edge to the active edge of the input signal to set the interrupt flag, when the lines are used as inputs.



PIA Read/Write (R/W) — This signal is generated by the MPU to control the direction of data transfers on the Data Bus. A low state on the PIA Read/Write line enables the input buffers and data is transferred from the MPU to the PIA on the E signal if the device has been selected. A high on the Read/Write line sets up the PIA for a transfer of data to the bus. The PIA output buffers are enabled when the proper address and the enable pulse E are present.

Reset — The active low $\overline{\text{Reset}}$ line is used to reset all register bits in the PIA to a logical zero (low). This line can be used as a power-on reset and as a master reset during system operation.

PIA Chip Select ($\overline{\text{CS0}}$, $\overline{\text{CS1}}$ and $\overline{\text{CS2}}$) — These three input signals are used to select the PIA. $\overline{\text{CS0}}$ and $\overline{\text{CS1}}$ must be high and $\overline{\text{CS2}}$ must be low for selection of the device. Data transfers are then performed under the control of the Enable and Read/Write signals. The chip select lines must be stable for the duration of the E pulse.

PIA Register Select ($\overline{\text{RS0}}$ and $\overline{\text{RS1}}$) — The two register select lines are used to select the various registers inside the PIA. These two lines are used in conjunction with internal Control Registers to select a particular register

that is to be written or read.

The Register select lines should be stable for the duration of the E pulse while in the read or write cycle.

Interrupt Request ($\overline{\text{IRQA}}$ and $\overline{\text{IRQB}}$) — The active low Interrupt Request lines ($\overline{\text{IRQA}}$ and $\overline{\text{IRQB}}$) act to interrupt the MPU either directly or through interrupt priority circuitry. These lines are "open source" (no load device on the chip) and are capable of sinking a current of 1.6 mA from an external source. This permits all interrupt request lines to be tied together in a wire-OR configuration.

Each Interrupt Request line has two internal interrupt flag bits that will cause the Interrupt Request line to go low. Each flag bit is associated with a particular peripheral interrupt line. Also four interrupt enable bits are provided in the PIA which may be used to inhibit a particular interrupt from a peripheral device.

Servicing an interrupt by the MPU may be accomplished by a software routine that, on a prioritized basis, sequentially reads and tests the two control registers in each PIA for interrupt flag bits that are set.

The Interrupt Flag is cleared (zeroed) as a result of an MPU Read Peripheral Data Operation.

PIA PERIPHERAL INTERFACE LINES

The PIA provides two 8-bit bi-directional data buses and four interrupt/control lines for interfacing to peripheral devices.

Section A Peripheral Data (PA0-PA7) — Each of the peripheral data lines can be programmed to act as an input or output. This is accomplished by setting a "1" in the corresponding Data Direction Register bit for those lines which are to be outputs. A "0" in a bit of the Data Direction Register causes the corresponding peripheral data line to act as an input. During an MPU Read Peripheral Data Operation, the data on peripheral lines programmed to act as inputs appears directly on the corresponding MPU Data Bus lines. In the input mode the internal pullup resistor on these lines represents a maximum of one standard TTL load.

The data in Output Register A will appear on the data lines that are programmed to be outputs. A logical "1" written into the register will cause a "high" on the corresponding data line while a "0" results in a "low". Data in Output Register A may be read by an MPU "Read Peripheral Data A" operation when the corresponding lines are programmed as outputs. This data will be read properly if the voltage on the peripheral data lines is greater than 2.0 volts for a logic "1" output and less than 0.8 volt for a logic "0" output. Loading the output lines such that the voltage on these lines does not reach full voltage causes the data transferred into the MPU on a Read operation to differ from that contained in the respective bit of Output Register A.

Section B Peripheral Data (PB0-PB7) — The peripheral data lines in the B Section of the PIA can be programmed

to act as either inputs or outputs in a similar manner to PA0-PA7. However, the output buffers driving these lines differ from those driving lines PA0-PA7. They have three-state capability, allowing them to enter a high impedance state when the peripheral data line is used as an input. In addition, data on the peripheral data lines PB0-PB7 will be read properly from those lines programmed as outputs even if the voltages are below 2.0 volts for a "high". As outputs, these lines are compatible with standard TTL and may also be used as a source of up to 1 milliampere at 1.5 volts to directly drive the base of a transistor switch.

Interrupt Input (CA1 and CB1) — Peripheral Input lines CA1 and CB1 are input only lines that set the interrupt flags of the control registers. The active transition for these signals is also programmed by the two control registers.

Peripheral Control (CA2) — The peripheral control line CA2 can be programmed to act as an interrupt input or as a peripheral control output. As an output, this line is compatible with standard TTL; as an input the internal pullup resistor on this line represents one standard TTL load. The function of this signal line is programmed with Control Register A.

Peripheral Control (CB2) — Peripheral Control line CB2 may also be programmed to act as an interrupt input or peripheral control output. As an input, this line has high input impedance and is compatible with standard TTL. As an output it is compatible with standard TTL and may also be used as a source of up to 1 milliampere at 1.5 volts to directly drive the base of a transistor switch. This line is programmed by Control Register B.



INTERNAL CONTROLS

There are six locations within the PIA accessible to the MPU data bus: two Peripheral Registers, two Data Direction Registers, and two Control Registers. Selection of these locations is controlled by the RS0 and RS1 inputs together with bit 2 in the Control Register, as shown in Table 1.

TABLE 1 – INTERNAL ADDRESSING

RS1	RS0	Control Register Bit		Location Selected
		CRA-2	CRB-2	
0	0	1	X	Peripheral Register A
0	0	0	X	Data Direction Register A
0	1	X	X	Control Register A
1	0	X	1	Peripheral Register B
1	0	X	0	Data Direction Register B
1	1	X	X	Control Register B

X = Don't Care

INITIALIZATION

A low reset line has the effect of zeroing all PIA registers. This will set PA0-PA7, PB0-PB7, CA2 and CB2 as inputs, and all interrupts disabled. The PIA must be configured during the restart program which follows the reset.

Details of possible configurations of the Data Direction and Control Register are as follows.

DATA DIRECTION REGISTERS (DDRA and DDRB)

The two Data Direction Registers allow the MPU to control the direction of data through each corresponding peripheral data line. A Data Direction Register bit set at "0" configures the corresponding peripheral data line as an input; a "1" results in an output.

CONTROL REGISTERS (CRA and CRB)

The two Control Registers (CRA and CRB) allow the MPU to control the operation of the four peripheral control lines CA1, CA2, CB1 and CB2. In addition they allow the MPU to enable the interrupt lines and monitor the status of the interrupt flags. Bits 0 through 5 of the two registers may be written or read by the MPU when the proper chip select and register select signals are applied. Bits 6 and 7 of the two registers are read only and are modified by external interrupts occurring on control lines CA1, CA2, CB1 or CB2. The format of the control words is shown in Table 2.

TABLE 2 – CONTROL WORD FORMAT

	7	6	5	4	3	2	1	0
CRA	IRQA1	IRQA2	CA2 Control			DDRA Access	CA1 Control	
CRB	IRQB1	IRQB2	CB2 Control			DDRB Access	CB1 Control	

Data Direction Access Control Bit (CRA-2 and CRB-2) –

Bit 2 in each Control register (CRA and CRB) allows selection of either a Peripheral Interface Register or the Data Direction Register when the proper register select signals are applied to RS0 and RS1.

Interrupt Flags (CRA-6, CRA-7, CRB-6, and CRB-7) –

The four interrupt flag bits are set by active transitions of signals on the four Interrupt and Peripheral Status lines when those lines are programmed to be inputs. These bits cannot be set directly from the MPU Data Bus and are reset indirectly by a Read Peripheral Data Operation on the appropriate section.

TABLE 3 – CONTROL OF INTERRUPT INPUTS CA1 AND CB1

CRA-1 (CRB-1)	CRA-0 (CRB-0)	Interrupt Input CA1 (CB1)	Interrupt Flag CRA-7 (CRB-7)	MPU Interrupt Request IRQA (IRQB)
0	0	↓ Active	Set high on ↓ of CA1 (CB1)	Disabled — IRQ remains high
0	1	↓ Active	Set high on ↓ of CA1 (CB1)	Goes low when the interrupt flag bit CRA-7 (CRB-7) goes high
1	0	↑ Active	Set high on ↑ of CA1 (CB1)	Disabled — IRQ remains high
1	1	↑ Active	Set high on ↑ of CA1 (CB1)	Goes low when the interrupt flag bit CRA-7 (CRB-7) goes high

- Notes:
- ↑ indicates positive transition (low to high)
 - ↓ indicates negative transition (high to low)
 - The Interrupt flag bit CRA-7 is cleared by an MPU Read of the A Data Register, and CRB-7 is cleared by an MPU Read of the B Data Register.
 - If CRA-0 (CRB-0) is low when an interrupt occurs (Interrupt disabled) and is later brought high, IRQA (IRQB) occurs on the positive transition of CRA-0 (CRB-0).



Control of CA1 and CB1 Interrupt Input Lines (CRA-0, CRB-0, CRA-1, and CRB-1) — The two lowest order bits of the control registers are used to control the interrupt input lines CA1 and CB1. Bits CRA-0 and CRB-0 are

used to enable the MPU interrupt signals IRQA and IRQB, respectively. Bits CRA-1 and CRB-1 determine the active transition of the interrupt input signals CA1 and CB1 (Table 3).

TABLE 4 — CONTROL OF CA2 AND CB2 AS INTERRUPT INPUTS
CRA5 (CRB5) is low

CRA-5 (CRB-5)	CRA-4 (CRB-4)	CRA-3 (CRB-3)	Interrupt Input CA2 (CB2)	Interrupt Flag CRA-6 (CRB-6)	MPU Interrupt Request IRQA (IRQB)
0	0	0	↓ Active	Set high on ↓ of CA2 (CB2)	Disabled — IRQ re- mains high
0	0	1	↓ Active	Set high on ↓ of CA2 (CB2)	Goes low when the interrupt flag bit CRA-6 (CRB-6) goes high
0	1	0	↑ Active	Set high on ↑ of CA2 (CB2)	Disabled — IRQ re- mains high
0	1	1	↑ Active	Set high on ↑ of CA2 (CB2)	Goes low when the interrupt flag bit CRA-6 (CRB-6) goes high

- Notes:
- ↑ indicates positive transition (low to high)
 - ↓ indicates negative transition (high to low)
 - The Interrupt flag bit CRA-6 is cleared by an MPU Read of the A Data Register and CRB-6 is cleared by an MPU Read of the B Data Register.
 - If CRA-3 (CRB-3) is low when an interrupt occurs (Interrupt disabled) and is later brought high, IRQA (IRQB) occurs on the positive transition of CRA-3 (CRB-3).

TABLE 5 — CONTROL OF CB2 AS AN OUTPUT
CRB-5 is high

CRB-5	CRB-4	CRB-3	CB2	
			Cleared	Set
1	0	0	Low on the positive transition of the first E pulse following an MPU Write "B" Data Register operation.	High when the interrupt flag bit CRB-7 is set by an active transition of the CB1 signal.
1	0	1	Low on the positive transition of the first E pulse following an MPU Write "B" Data Register operation.	High on the positive transition of the next "E" pulse.
1	1	0	Low when CRB-3 goes low as a result of an MPU Write in Control Register "B".	Always low as long as CRB-3 is low. Will go high on an MPU Write in Control Register "B" that changes CRB-3 to "one".
1	1	1	Always high as long as CRB-3 is high. Will be cleared when an MPU Write Control Register "B" results in clearing CRB-3 to "zero".	High when CRB-3 goes high as a result of an MPU write into control register "B".



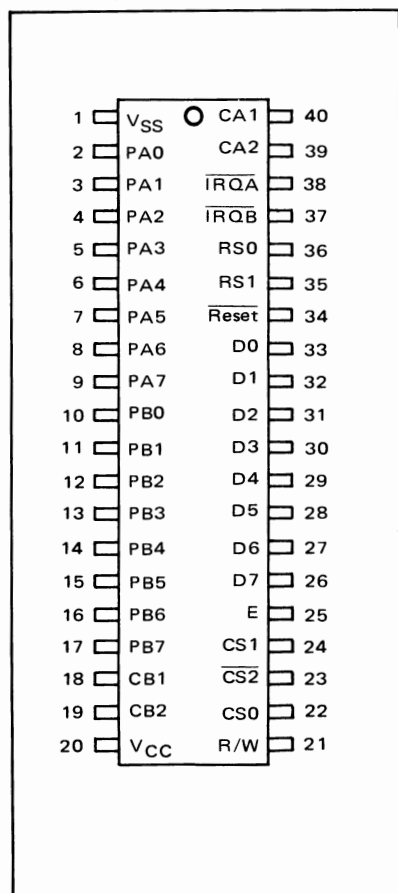
Control of CA2 and CB2 Peripheral Control Lines (CRA-3, CRA-4, CRA-5, CRB-3, CRB-4, and CRB-5) — Bits 3, 4, and 5 of the two control registers are used to control the CA2 and CB2 Peripheral Control lines. These bits determine if the control lines will be an interrupt input or an output control signal. If bit CRA-5 (CRB-5)

is low, CA2 (CB2) is an interrupt input line similar to CA1 (CB1) (Table 4). When CRA-5 (CRB-5) is high, CA2 (CB2) becomes an output signal that may be used to control peripheral data transfers. When in the output mode, CA2 and CB2 have slightly different characteristics (Tables 5 and 6).

TABLE 6 — CONTROL OF CA2 AS AN OUTPUT
CRA-5 is high

CRA-5	CRA-4	CRA-3	CA2	
			Cleared	Set
1	0	0	Low on negative transition of E after an MPU Read "A" Data operation.	High on an active transition of the CA1 signal.
1	0	1	Low immediately after an MPU Read "A" Data operation.	High on the negative edge of the next "E" pulse.
1	1	0	Low when CRA-3 goes low as a result of an MPU Write in Control Register "A".	Always low as long as CRA-3 is low.
1	1	1	Always high as long as CRA-3 is high.	High when CRA-3 goes high as a result of a Write in Control Register "A".

PIN ASSIGNMENT



PACKAGE DIMENSIONS

